

D319

AWS Cloud Architecture

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

An online pharmacy with 600,000 customers runs its order database on Amazon RDS for MySQL in us-east-1. The CTO, who joined from a telecom company, requires the database to keep running automatically if its Availability Zone fails. What should the architect configure?

- A. RDS Multi-AZ deployment
- B. A larger DB instance class
- C. An RDS read replica in the same AZ
- D. Nightly snapshots to S3 only

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Heavy reporting queries slow down a production RDS database. Which change offloads them?

- A. Create a read replica and point reports at it.
- B. Take more frequent snapshots.
- C. Enable Multi-AZ.
- D. Move the database to Glacier.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A law firm with 90 attorneys must keep closed case files for 10 years for compliance. The files are almost never read, and a retrieval time of up to 48 hours is acceptable. The firm uses the us-west-2 region. Which S3 storage class is the lowest-cost fit?

- A. S3 Standard-IA
- B. S3 Intelligent-Tiering
- C. S3 Standard
- D. S3 Glacier Deep Archive

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A team stores backups that are read about once a month but must be available within milliseconds when needed. Which S3 class fits?

- A. S3 Glacier Deep Archive
- B. S3 Standard
- C. Any Glacier class, since all Glacier retrievals are instant
- D. S3 Standard-IA

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A retail web tier in a VPC with three public subnets is being hit by requests from one malicious IP address. The security team has 11 members on call. The architect must block that single address from reaching the subnet. What should be used?

- A. Remove the instances' IAM role
- B. A security group deny rule
- C. A network ACL deny rule for that IP
- D. Add three more subnets

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A NACL allows inbound HTTPS on port 443, but clients time out. The security group allows 443 in. What is the likely missing rule?

- A. An inbound rule for port 80
- B. An outbound security group rule for the return ports
- C. Nothing; NACLs track connections like security groups.
- D. An outbound NACL rule for the ephemeral return ports

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

An image-processing app on EC2, written by a contractor in Python, must write results to an S3 bucket. The contractor put an access key and secret key in the config file. The app processes 50,000 images a day. What is the best practice?

- A. Give the role full S3 access to avoid permission errors.
- B. Keep the keys but rotate them yearly.
- C. Attach an IAM role to the instance with write access to only that bucket, and remove the keys.
- D. Encrypt the config file and keep the root account keys in it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A ticketing site's web tier, which uses a React front end, accepts orders much faster than the payment workers can process them during flash sales. Orders are lost when workers are overwhelmed. The site runs in two Regions. What design fixes this?

- A. Make the web tier call workers directly with longer timeouts.
- B. Put orders in an Amazon SQS queue that the workers poll.
- C. Add a third Region.
- D. Publish orders to an SNS topic.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A company with 12 microservices built by four teams wants one load balancer that sends /api/orders to one target group and /api/users to another. Which load balancer should it use?

- A. Application Load Balancer
- B. 12 separate load balancers, one per service
- C. Network Load Balancer
- D. An internet gateway

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A news site with readers on five continents stores images and static pages in S3. Readers in Asia complain about slow page loads. The editorial team publishes 200 articles a day. What most directly speeds delivery to those readers?

- A. Add ElastiCache in front of the database.
- B. Publish fewer articles a day.
- C. Serve the content through Amazon CloudFront.
- D. Move the bucket to Glacier.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A research lab runs a genome analysis batch that can stop and resume from checkpoints at any time. The lab's director won a grant last year. The job runs for a few days each month. Which EC2 purchasing option costs the least?

- A. Reserved Instances for three years
- B. Spot Instances
- C. On-Demand Instances
- D. Dedicated Hosts

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Which workload is the best fit for a one-year Reserved Instance or Savings Plan?

- A. A batch job that can be stopped at any time
- B. A production web server that runs 24 hours a day all year
- C. A workload whose future size is unknown
- D. A test job run for two days next month

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

An online pharmacy ~~with 600,000 customers~~ runs its order database on Amazon RDS for MySQL in us-east-1. The CTO, ~~who joined from a telecom company~~, requires the database to keep running automatically if its Availability Zone fails. What should the architect configure?

Target word: HA for RDS **Decides:** Automatic failover on AZ loss.

A	Correct	A synchronous standby in another AZ takes over automatically.
B	Adjacent category	More power, still one AZ.
C	Confusion set	Read replicas scale reads and share the AZ failure.
D	Dropped element	Allows recovery but not automatic failover.

2. Answer A Concept 1 · transfer

Heavy reporting queries slow down a production RDS database. Which change offloads them?

Target word: scale reads **Decides:** Read traffic needs to move off the primary.

A	Correct	Replicas serve read-only traffic.
B	Adjacent category	Backups do not serve queries.
C	Inverted operation	The standby does not serve reads; it is for failover.
D	Confusion set	Archive storage, not a database.

3. Answer D Concept 2

A law firm ~~with 90 attorneys~~ must keep closed case files for 10 years for compliance. The files are almost never read, and a retrieval time of up to 48 hours is acceptable. ~~The firm uses the us-west-2 region.~~ Which S3 storage class is the lowest-cost fit?

Target word: cheapest archive class **Decides:** Almost never read; 48 hours is acceptable.

A	Adjacent category	Infrequent access with fast retrieval, costing more than archive.
B	Confusion set	Best when access patterns are unknown; here they are known.
C	Inverted operation	Priced for frequent access.
D	Correct	Lowest cost, with retrieval within about 12 to 48 hours.

4. Answer D Concept 2 · transfer

A team stores backups that are read about once a month but must be available within milliseconds when needed. Which S3 class fits?

Target word: class for infrequent but instant access **Decides:** Infrequent reads plus millisecond retrieval.

A	Inverted operation	Retrieval takes hours.
B	Adjacent category	Works but costs more for monthly access.
C	Absolute language	Only Glacier Instant Retrieval is immediate; others take minutes to hours.
D	Correct	Lower storage cost, immediate access.

5. Answer C Concept 3

A retail web tier in a VPC with three public subnets is being hit by requests from one malicious IP address. The security team has 11 members on call. The architect must block that single address from reaching the subnet. What should be used?

Target word: control that can deny one IP **Decides:** Explicit deny is needed.

A	Adjacent category	IAM controls API permissions, not network traffic.
B	Confusion set	Security groups allow only; they have no deny rules.
C	Correct	NACLs support explicit deny at the subnet.
D	Camouflage	Subnet count does not block traffic.

6. Answer D Concept 3 · transfer

A NACL allows inbound HTTPS on port 443, but clients time out. The security group allows 443 in. What is the likely missing rule?

Target word: stateless return traffic **Decides:** NACLs do not remember connections.

A	Adjacent category	The traffic is HTTPS; port 80 does not fix replies.
B	Inverted operation	Security groups are stateful; replies are allowed automatically.
C	Confusion set	NACLs are stateless.
D	Correct	Stateless NACLs need explicit rules for responses.

7. Answer C Concept 4

An image-processing app on EC2, written by a contractor in Python, must write results to an S3 bucket. The contractor put an access key and secret key in the config file. The app processes 50,000 images a day. What is the best practice?

Target word: how an EC2 app gets AWS permissions **Decides:** Workloads on AWS should use roles, not stored keys.

A	Absolute language	Violates least privilege.
B	Dropped element	Long-term keys on a server remain a leak risk.
C	Correct	Temporary role credentials plus least privilege.
D	Inverted operation	Root keys are the highest risk possible.

8. Answer B Concept 5

A ticketing site's web tier, which uses a React front end, accepts orders much faster than the payment workers can process them during flash sales. Orders are lost when workers are overwhelmed. The site runs in two Regions. What design fixes this?

Target word: decoupling to absorb spikes **Decides:** Work must wait safely until a worker is free.

A	Inverted operation	Keeps the tight coupling that causes the loss.
B	Correct	The queue buffers orders so none are lost.
C	Camouflage	More Regions do not buffer work between tiers.
D	Confusion set	SNS pushes to subscribers and does not hold work for busy workers.

9. Answer A Concept 6

A company with 12 microservices ~~built by four teams~~ wants one load balancer that sends /api/orders to one target group and /api/users to another. Which load balancer should it use?

Target word: load balancer for path routing **Decides:** Routing by URL path is a layer 7 feature.

A	Correct	ALB supports path-based routing rules.
B	Wrong stem number	Works but ignores that one ALB can route by path.
C	Confusion set	Layer 4; it does not read URL paths.
D	Adjacent category	Connects a VPC to the internet; it does not balance load.

10. Answer C Concept 7

A news site with readers on five continents stores images and static pages in S3. Readers in Asia complain about slow page loads. ~~The editorial team publishes 200 articles a day.~~ What most directly speeds delivery to those readers?

Target word: cache near distant users **Decides:** Latency comes from distance to the origin.

A	Adjacent category	Speeds database reads, not delivery of static files to far users.
B	Camouflage	Article volume is not the cause.
C	Correct	Edge locations cache content close to readers.
D	Inverted operation	Makes retrieval slower.

11. Answer B Concept 8

A research lab runs a genome analysis batch that can stop and resume from checkpoints at any time. ~~The lab's director won a grant last year.~~ The job runs for a few days each month. Which EC2 purchasing option costs the least?

Target word: cheapest option for interruptible work **Decides:** The job tolerates interruption.

A	Confusion set	Commits to steady use; the job runs only a few days a month.
B	Correct	Deep discounts for workloads that can handle interruptions.
C	Adjacent category	Works but costs more than Spot.
D	Inverted operation	The most expensive option.

12. Answer B Concept 8 · transfer

Which workload is the best fit for a one-year Reserved Instance or Savings Plan?

Target word: reverse: match workload to a commitment **Decides:** Steady, always-on use.

A	Adjacent category	Better suited to Spot.
B	Correct	Steady use earns the full discount.
C	Confusion set	On-Demand fits unpredictable needs.
D	Inverted operation	Short use wastes a one-year commitment.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	10	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Multi-AZ versus read replicas	RDS Multi-AZ keeps a synchronous standby for automatic failover (availability); read replicas copy data asynchronously to spread read traffic (performance).	Using a read replica as the high-availability answer, or Multi-AZ to scale reads.	"Survive an AZ outage" versus "reports slow down the database".
2	S3 storage classes	Standard for frequent access; Standard-IA for infrequent access needing fast retrieval; Glacier classes for archives with slower retrieval; Intelligent-Tiering when access patterns are unknown.	Putting data needed within milliseconds into Glacier Deep Archive.	How often data is read and how fast it is needed.
3	Security groups versus network ACLs	Security groups are stateful, instance-level, allow-only; network ACLs are stateless, subnet-level, and support explicit deny rules evaluated in number order.	Trying to block one IP with a security group, or forgetting return traffic on a NACL.	Need to deny a specific address or a note about return traffic.
4	IAM roles for AWS workloads	Give EC2 instances and Lambda functions IAM roles for temporary credentials; never store long-term access keys on servers or in code; grant least privilege.	Hard-coding an access key in the app config.	An application on AWS needs to call another AWS service.
5	Decoupling with queues and topics	SQS buffers work between producers and consumers so spikes do not overload the back end; SNS fans a message out to many subscribers.	Using SNS when messages must wait for a busy worker.	Front end is faster than back end, or one event needs many receivers.
6	Load balancer choice	Application Load Balancer routes at layer 7 by path or host; Network Load Balancer handles layer 4 TCP/UDP with very high throughput and static IPs.	Choosing NLB for URL path routing.	Path-based routing versus raw TCP performance.
7	Caching layers	CloudFront caches content at edge locations close to users; ElastiCache holds frequent query results in memory to offload a database.	Using CloudFront to reduce repeated database queries from the app tier.	Far-away users versus a hot database.
8	EC2 purchasing options	Reserved Instances or Savings Plans for steady use; Spot for fault-tolerant, interruptible work; On-Demand for short or unpredictable workloads.	Buying a reservation for a job that runs a few days a year.	Workload pattern and tolerance for interruption.

Course scope (public description): AWS Cloud Architecture examines the skills and knowledge needed to effectively design structured cloud environments. Through practical application, students will gain experience in

designing control measures for resilient architectures with cloud solutions and concepts, and to design high-performing and scalable architectures for software performance workloads. Students will also learn skills in designing security policies and access for cloud applications and architectures, and designing cost optimized storage, database and network architectures based on situational feedback.